CUNEF UNIVERSIDAD

Econometrics 2024/2025

Topic 1: Simple Regression Model

Outline

The simple regression model

Deriving the ordinary least squares (OLS) estimates

Properties of OLS

Goodness of fit

Incorporating non-linearities

Expected values and variances of the OLS estimators

Regression on a binary explanatory variable

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The simple regression model

- We are often interested in "explaining y in terms of x", or in studying "how y changes when x changes".
- This can be achieved with the simple regression model.

Intercept Slope parameter

$$y = \beta_0 + \beta_1 x + u$$

Error term, disturbance, unobservables, ...

Dependent variable, explained variable, response variable, ...

Independent variable, explanatory variable, regressor, ...

The simple regression model

Interpretation of the simple linear regression model:

$$\frac{\Delta y}{\Delta x} = \beta_1 \quad \text{as long as} \quad \frac{\Delta u}{\Delta x} = 0$$

By how much does the dependent variable change if the independent variable is increased by one unit?

Interpretation only correct if all other things remain equal when the independent variable is increased by one unit

The Simple Regression Model

- **Example: CAPM equation**

$$R_i - R_f = \beta_0 + \beta_1[Rm - R_f] + u$$

Other risk factors

Measures the sensitivity of stock i to market changes

- **Example: wage equation**

$$wage = \beta_0 + \beta_1\ education + u$$

Labor force experience, tenure with current employer, work ethic, intelligence, ...

Measures the change in hourly wage given another year of education

The simple regression model

- Conditional mean independence assumption:

$$E(u \mid x) = 0$$

The explanatory variable must not contain information about the mean of the unobserved factors.

- Example: wage equation

$$wage = \beta_0 + \beta_1 \; education + u \longrightarrow \text{Intelligence} \ldots$$

The conditional mean independence assumption is unlikely to hold because individuals with more education will also be more intelligent on average.

The simple regression model

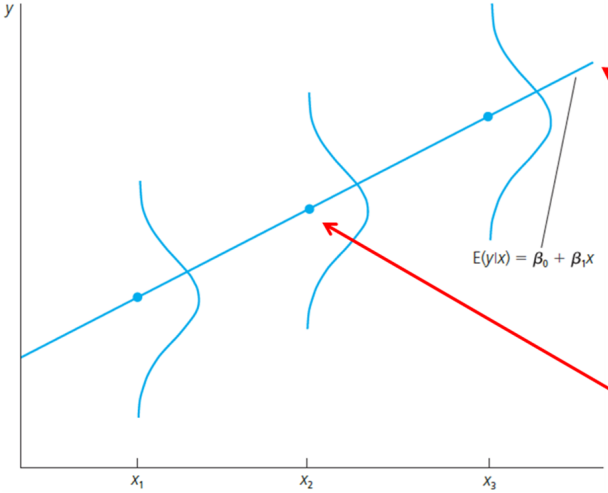
Population regression function (PFR)

- The conditional mean independence assumption implies that

$$\begin{aligned} E(y \mid x) &= E\left(\beta_0 + \beta_1 x + u \mid x\right) \\ &= \beta_0 + \beta_1 x + E(u \mid x) \\ &= \beta_0 + \beta_1 x \end{aligned}$$

- This means that the average value of the dependent variable can be expressed as a linear function of the explanatory variable.

The simple regression model



Population regression function

For individuals with $x = x_2$, the average value of y is $\beta_0 + \beta_1 x_2$

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Deriving the ordinary least squares (OLS) estimates

In order to estimate the regression model one needs random sample of n observations

$$
\begin{aligned}
(x_1, y_1) &\longleftarrow \text{First observation} \\
(x_2, y_2) &\longleftarrow \text{Second observation} \\
(x_3, y_3) &\longleftarrow \text{Third observation} \\
&\vdots \\
(x_n, y_n) &\longleftarrow \text{n-th observation}
\end{aligned}
$$

Deriving the ordinary least squares (OLS) estimates

- Defining regression residuals

$$\widehat{u}_i = y_i - \widehat{y}_i = y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i$$

- Minimize the sum of the squared regression residuals (SSR)

$$\min \text{SSR} = \min \sum_{i=1}^{n} \widehat{u}_i^2 \rightarrow \widehat{\beta}_0, \widehat{\beta}_1$$

- OLS estimators

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}, \quad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$$

Deriving the ordinary least squares estimates

- Where do these expressions come from?
- Taking the first-order derivative of $\sum_{i=1}^{n} \widehat{u}_i^2$ with respect to β_0 and β_1 yields:

$$\frac{\partial SSR}{\partial \beta_0} = -2 \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i) = 0$$
$$\frac{\partial SSR}{\partial \beta_1} = -2 \sum_{i=1}^{n} x_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

- Algebraic manipulation of these two equations leads to the OLS expressions for $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

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Properties of OLS on any sample of data

- Fitted (predicted) values

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$$

- Residuals (deviations from the regression line)

$$\widehat{u}_i = y_i - \widehat{y}_i$$

- Deviations from regression line sum up to zero:

$$\sum_{i=1}^{n} \widehat{u}_i = 0$$

- Covariance between deviations and regressors is zero

$$\sum_{i=1}^{n} x_i \widehat{u}_i = 0$$

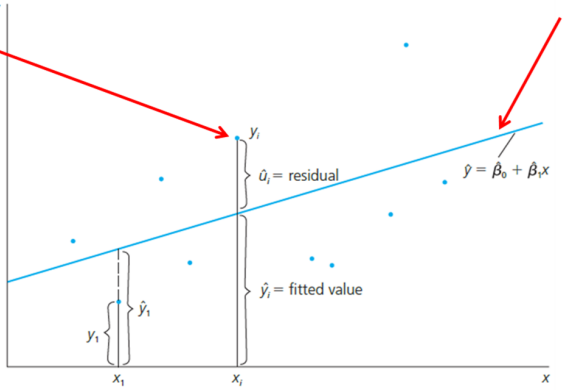
- Sample averages of y and x lie on regression line

$$\bar{y} = \widehat{\beta}_0 + \widehat{\beta}_1 \bar{x}$$

OLS fits as good as possible a regression line through the data points

For example, the i-th data point (x_i, y_i)

Fitted regression line



Example of a simple regression

- CEO salary and return on equity

$$\text{CEO salary} = \beta_0 + \beta_1 \text{ROE} + u$$

Salary in thousands of dollars

Average return on equity on CEO's firm

- Fitted regression:

$$\widehat{\text{CEO salary}} = 963.191 + 18.501\text{ROE}$$

If the return on equity increases by 1 percent, then salary is predicted to change by $\$18,501$

Example of a simple regression

obsno	roe	salary	salaryhat	uhat
1	14.1	1095	1224.058	-129.058
2	10.9	1001	1164.854	-163.854
3	23.5	1122	1397.960	-275.969
4	5.9	578	1072.348	-494.348
5	13.8	1368	1218.508	149.493
6	20.0	1145	1333.215	-188.215
7	16.4	1078	1266.611	188.611
8	16.3	1094	1264.761	-170.761
9	10.5	1237	1157.454	79.546
10	26.3	833	1449.773	-616.773
11	25.9	567	1442.372	-875.372
12	26.8	933	1459.023	-526.023
13	14.8	1339	1237.009	101.991
14	22.3	937	1375.768	-438.768
15	56.3	2011	2004.808	6.192

- This table presents fitted values and residuals for 15 CEOs.
- For example, the 12th CEO's predicted salary is $\$526,023$ higher than their actual salary.
- By contrast the 5th CEO's predicted salary is $\$149,493$ lower than their actual salary.

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Goodness of fit: How well does an explanatory variable explain the dependent variable?

- **Total sum of squares**: represents total variation in the dependent variable

$$SST \equiv \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- **Explained sum of squares**: represents variation explained by regression

$$SSE \equiv \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2$$

- **Residual sum of squares**: represents variation not explained by regression

$$SSR \equiv \sum_{i=1}^{n} \hat{u}_i{}^2$$

Goodness of fit: How well does an explanatory variable explain the dependent variable?

- Decomposition of total variation:

$$\text{Total variation (SST)} = \text{Explained part (SSE)} + \text{Unexplained part (SSR)}$$

- Goodness-of-fit measure (R-squared)

$$R^2 \equiv \frac{SSE}{SST} = 1 - \frac{SSR}{SST}$$

R-squared measures the fraction of the total variation that is explained by the regression

Goodness of fit: How well does an explanatory variable explain the dependent variable?

- CEO salary and return on equity

$$\widehat{\text{CEO salary}} = 963.191 + 18.501\text{ROE}$$
$$n = 209, \quad R^2 = 0.0132$$

The regression explains only 1.3% of the total variation in salaries.

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Incorporating non-linearities: Semi-logarithmic form

- Regression of log wages on years of education

$$\log(\text{ wage }) = \beta_0 + \beta_1 educ + u$$

- This changes the interpretation of the regression coefficient:

$$\beta_1 = \frac{\Delta \log(\text{ wage })}{\Delta \text{ educ}} = \frac{1}{\text{wage}} \cdot \frac{\Delta \text{ wage}}{\Delta \text{ educ}} = \frac{\frac{\Delta \text{ wage}}{\text{wage}}}{\Delta \text{ educ}}$$

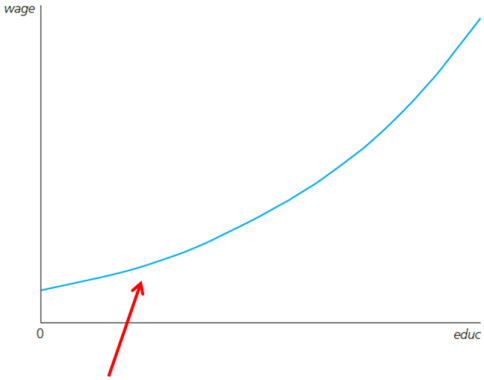
Now β_1 measures the **percentage change of wage** if **years of education are increased by one year**.

Incorporating non-linearities: Semi-logarithmic form

- Fitted regression:

  $\widehat{\log}(\text{ wage }) = 0.584 + 0.083\text{educ}$

  The wage increases by 8.3% for every additional year of education (= return to another year of education)



Incorporating nonlinearities: Log-logarithmic form

- Natural log of CEO salary and natural log of his/her firm sales:

$$\log(\text{ salary }) = \beta_0 + \beta_1 \log(\text{ sales }) + u$$

- This changes the interpretation of the regression coefficient:

$$\beta_1 = \frac{\Delta \log(\text{salary})}{\Delta \log(\text{sales})} = \frac{\frac{\Delta \text{salary}}{\text{salary}}}{\frac{\Delta \text{sales}}{\text{sales}}}$$

  - **Percentage** change in salary if sales increase by 1%.
  - Logarithmic changes are always percentage changes.

Incorporating nonlinearities: Log-logarithmic form

- CEO salary and firm sales: fitted regression

$$\widehat{\log(\text{salary})} = 4.822 + 0.257 \log(\text{sales})$$

  - $+1\%$ sales $\rightarrow$ $+0.257\%$ salary

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Expected values and variances of the OLS estimators

- The estimated regression coefficients are **random variables** because they are calculated from a **random sample**

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$$

  Data is random and depends on particular sample that has been drawn

- The question is: What the estimators will estimate on average and how large will their variability be in repeated samples

$$E\left(\widehat{\beta}_0\right) = ?, E\left(\widehat{\beta}_1\right) = ? \quad \text{Var}\left(\widehat{\beta}_0\right) = ?, \text{Var}\left(\widehat{\beta}_1\right) = ?$$

Standard assumptions for the linear regression model

- Assumption SLR. 1 (Linear in parameters)

$$y = \beta_0 + \beta_1 x + u \longleftarrow$$ In the population, the relationship between y and x is linear

- Assumption SLR. 2 (Random sampling)

$$\{(x_i, y_i) : i = 1, \ldots, n\} \longleftarrow$$ The data is a random sample drawn from the population

$$y_i = \beta_0 + \beta_1 x_i + u_i \longleftarrow$$ Each data point therefore follows the population equation

Standard assumptions for the linear regression model

- Assumption SLR. 3 (Sample variation in the explanatory variable)

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 > 0$$

  The values of the explanatory variables are not all the same (otherwise it would be impossible to study how different values of the explanatory variable lead to different values of the dependent variable)

- Assumption SLR. 4 (Zero conditional mean)

$$E\left(u_i \mid x_i\right) = 0$$

  The value of the explanatory variable must contain no information about the mean of the unobserved factors

Unbiasedness of OLS

- Under assumptions SLR1 – SLR4, we can show that:

$$E\left(\widehat{\beta}_0\right) = \beta_0, E\left(\widehat{\beta}_1\right) = \beta_1$$

- Interpretation of unbiasedness:
  - The estimated coefficients may be smaller or larger, depending on the sample that is the result of a random draw.
  - However, on average, they will be equal to the values that characterize the true relationship between y and x in the population.
  - "On average" means if sampling was repeated, i.e. if drawing the random sample and doing the estimation was repeated many times.
  - In a given sample, estimates may differ considerably from true values.

Variances of the OLS estimators

- Depending on the sample, the estimates will be nearer or farther away from the true population values.
- How far can we expect our estimates to be away from the true population values on average (= sampling variability)?
- Sampling variability is measured by the estimator's variances

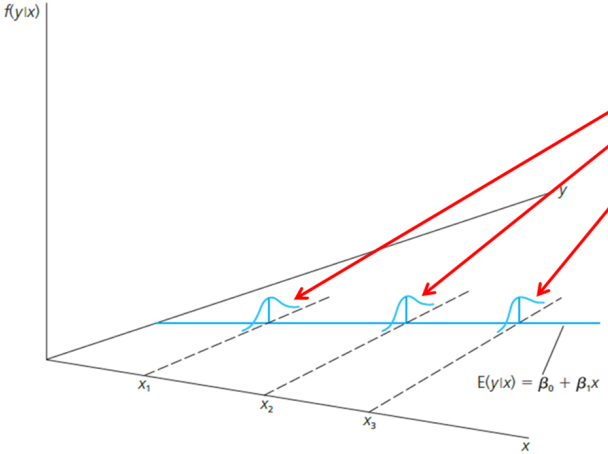
$$\text{Var}\left(\widehat{\beta}_0\right), \text{Var}\left(\widehat{\beta}_1\right)$$

- Assumption SLR. 5 (Homoskedasticity)

$$\text{Var}\left(u_i \mid x_i\right) = \sigma^2 \longleftarrow$$

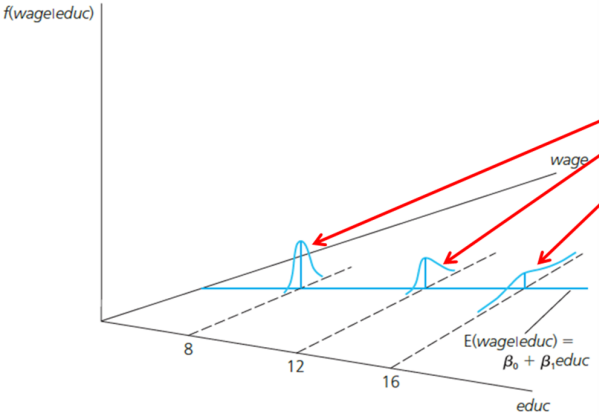
The value of the explanatory variable must contain no information about the variability of the unobserved factors

Graphical illustration of homoskedasticity



The variability of the unobserved influences does not depend on the value of the explanatory variable

An example for heteroskedasticity: Wage and education



The variance of the unobserved determinants of wages increases with the level of education

Variances of the OLS estimators

- Under assumptions SLR. 1 – SLR.5:

$$\mathrm{Var}\left(\widehat{\beta}_1\right) = \frac{\sigma^2}{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2} = \frac{\sigma^2}{SST_x}$$

$$\mathrm{Var}\left(\widehat{\beta}_0\right) = \frac{\sigma^2 n^{-1}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)^2} = \frac{\sigma^2 n^{-1}\sum_{i=1}^{n} x_i^2}{SST_x}$$

- Conclusion: The sampling variability of the estimated regression coefficients will be the higher, the larger the variability of the unobserved factors, and the lower, the higher the variation in the explanatory variable.

Estimating the error variance

- $$\mathrm{Var}\left(u_i \mid x_i\right) = \sigma^2 = \mathrm{Var}\left(u_i\right) \longleftarrow$$ The variance of u does not depend on x,

- $$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{u}_i - \overline{\hat{u}}_i\right)^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{u}_i^2 \longleftarrow$$ One could estimate the variance of the errors by calculating the variance of the residuals in the sample; unfortunately this estimate would be biased

- $$\widehat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}\widehat{u}_i^2 \longleftarrow$$ An unbiased estimate of the error variance can be obtained by substracting the number of estimated regression coefficients from the number of observations

Calculation of standard errors for regression coefficients

$$\text{se}\left(\widehat{\beta}_1\right) = \sqrt{\widehat{\text{Var}}\left(\widehat{\beta}_1\right)} = \sqrt{\widehat{\sigma}^2 / SST_x}$$

$$\text{se}\left(\hat{\beta}_0\right) = \sqrt{\widehat{\text{Var}}\left(\hat{\beta}_0\right)} = \sqrt{\hat{\sigma}^2 n^{-1} \sum_{i=1}^{n} x_i^2 / SST_x}$$

The estimated standard deviations of the regression coefficients are called "standard errors." They measure how precisely the regression coefficients are estimated.

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Regression on a binary explanatory variable

- Suppose that x is either equal to 0 or 1

$$y = \beta_0 + \beta_1 x + u$$
$$E(y \mid x = 0) = \beta_0 \quad E(y \mid x = 1) = \beta_0 + \beta_1$$

- This regression allows the mean value of y to differ depending on the state of x

$$\beta_1 = E(y \mid x = 1) - E(y \mid x = 0)$$

- Note that the statistical properties of OLS are no different when x is binary

Regression on a binary explanatory variable

- Example: The effects of a job training program on earnings
- Real earnings are regressed on a binary variable indicating participation in a job training program.

$$\widehat{re78} = 4.55 + 1.79 \text{ train}$$

$$n = 445, R^2 = 0.018$$

  re78 is real earnings in 1978 measured in thousands of dollars. train is a binary variable equal to 1 if the individual participated in the job training program
- Those who participated in the training program have earnings $\$1,790$ higher than those who did not participate.
- This represents a 39.3% increase over the $\$4,550$ average earnings from those who did not participate.

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